

Precision Drone Flight Control and Landing

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1. Introduction

Modern multirotor drones rely on carefully tuned feedback control systems to maintain stable flight and perform autonomous operations. Improper controller gains can lead to excessive oscillations, poor responsiveness, unstable hover, and unreliable navigation.

In this project, a custom PX4 Software-In-The-Loop (SITL) simulation running in Gazebo was provided with intentionally unstable PID parameters. The primary objective was to stabilize the drone through systematic PID tuning and subsequently develop a MAVSDK-based Python script capable of autonomously taking off, navigating to a designated helipad, and performing a safe landing without manual intervention.

This report summarizes the PID tuning process, the parameters modified, the challenges encountered during development, and the final autonomous landing approach.

2. Initial Drone Behaviour

The provided drone exhibited unstable flight characteristics immediately after takeoff. Significant roll and pitch oscillations prevented the vehicle from maintaining a steady hover, making autonomous navigation impractical.

The observed issues included:

- Continuous oscillations during hover
- Poor attitude stability
- Excessive drift
- Unpredictable response to control inputs

These behaviours indicated that the default controller gains required tuning before autonomous flight could be attempted.

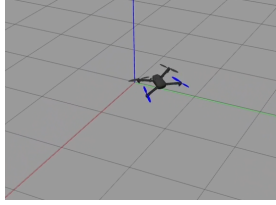


Figure 1: Initial Drone Behaviour

3. PID Tuning Process

The provided PID tuning GUI was used to iteratively modify the PX4 multicopter controller parameters while observing the drone's behaviour in simulation. Since the GUI did not expose the 'MC_ROLL_P' and 'MC_PITCH_P' attitude gain parameters, these were changed directly via PX4 parameter commands.

3.1 PID Tuning Workflow

The following workflow was repeated throughout the tuning process:

1. Launch the PX4 SITL simulation.
2. Connect the tuning GUI to the simulated drone.
3. Arm the vehicle.
4. Perform autonomous takeoff.
5. Observe hover behaviour.
6. Modify PID parameters.
7. Apply the updated values.
8. Repeat until satisfactory stability was achieved.

3.2 Tuning Strategy

Instead of modifying multiple parameters simultaneously, controller gains were adjusted incrementally while monitoring their individual effects on flight behaviour.

Particular attention was given to:

- Hover stability
- Oscillation reduction
- Responsiveness
- Position drift
- Smooth attitude recovery

Several iterations were performed before obtaining a balanced set of controller gains that provided stable hover while maintaining sufficient responsiveness for autonomous flight.

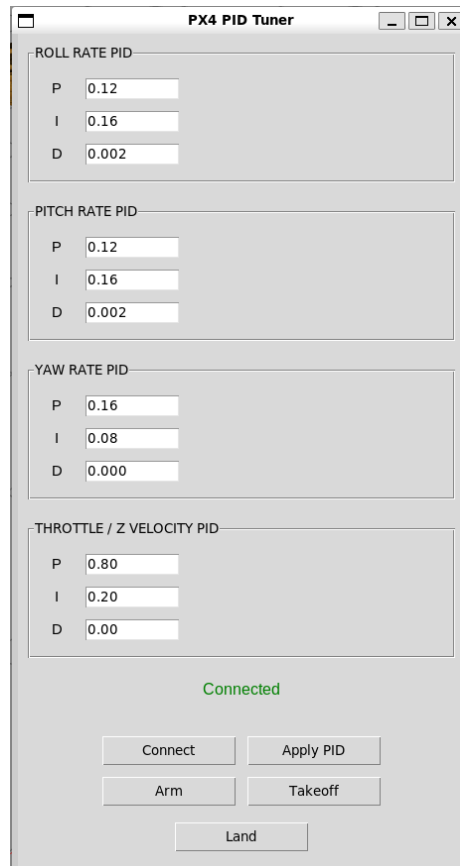


Figure 2: PID Tuner GUI

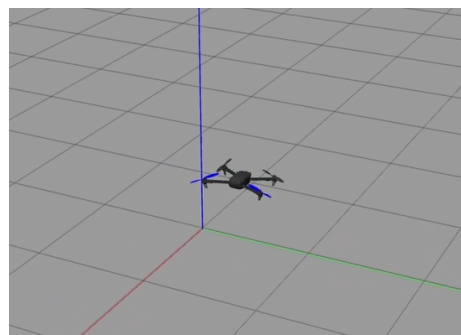


Figure 3: Stable Hover After PID Tuning

4. Parameters Modified

The following PX4 multicopter PID parameters were modified during the tuning process.

Parameter	Default	Final	Observation
MC_ROLL_P	0.01	6.50	Improved roll attitude stability
MC_PITCH_P	0.00	6.50	Reduced pitch oscillations
MC_YAW_P	2.80	2.80	No modification required
MC_ROLLRATE_P	0.12	0.15	Faster roll response
MC_ROLLRATE_I	0.16	0.20	Reduced steady-state error
MC_ROLLRATE_D	0.002	0.003	Increased damping and reduced oscillations
MC_PITCHRATE_P	0.12	0.15	Improved pitch response
MC_PITCHRATE_I	0.16	0.20	Improved hover stability
MC_PITCHRATE_D	0.002	0.003	Reduced pitch overshoot
MC_YAWRATE_P	0.16	0.20	Improved yaw responsiveness
MC_YAWRATE_I	0.08	0.10	Better heading correction
MC_YAWRATE_D	0.00	0.00	No modification required

The final parameter set significantly reduced oscillations and enabled the drone to achieve a stable hover suitable for autonomous navigation and landing.

5. Autonomous Landing Implementation

Once stable flight was achieved, a MAVSDK Python script was developed to automate the complete landing sequence.

The implemented workflow is summarized below:

1. Connect to PX4 SITL.
2. Wait for successful connection.
3. Arm the drone.
4. Perform autonomous takeoff.
5. Wait until the desired altitude is reached.
6. Enter Offboard mode.
7. Navigate to the landing waypoint.
8. Descend vertically.
9. Execute the PX4 landing command.
10. Wait until landing is complete.

The use of Offboard mode enabled precise waypoint navigation while the PX4 landing controller handled the final descent.

6. Challenges Faced

Several technical challenges were encountered during development.

6.1 PID Instability

The initial controller gains caused severe oscillations immediately after takeoff, preventing stable hover.

Repeated tuning and testing were required to balance controller responsiveness and damping.

6.2 Hover Stability

Even after reducing oscillations, maintaining a consistent hover required further adjustment of the rate controllers to minimize drift while preserving smooth control response.

6.3 Non-Editable Attitude Gain Parameters

During the PID tuning process, it was observed that the provided tuning GUI did not expose the attitude controller gains 'MC_ROLL_P' and 'MC_PITCH_P' for modification. The values were therefore changed manually using PX4 parameter commands and documented accordingly.

This additional tuning step significantly improved hover stability and reduced the large oscillations exhibited by the drone during takeoff and hover.

6.4 Offboard Mode Initialization

PX4 requires valid position setpoints to be published before enabling Offboard mode. Attempting to switch modes prematurely resulted in Offboard activation failures.

This issue was resolved by sending initial position setpoints prior to starting Offboard mode.

6.5 Coordinate Frame Calibration

Although the challenge specifies the helipad location at world coordinates **(5, 3)**, the local coordinate frame used by MAVSDK differed from the Gazebo world frame.

Experimental calibration showed that commanding the drone to the local waypoint **(2, 4)** consistently aligned the vehicle above the designated helipad. Therefore, the calibrated waypoint was used in the autonomous landing script to achieve reliable landings.

7. Final Approach Used

The completed autonomous landing algorithm follows the sequence below:

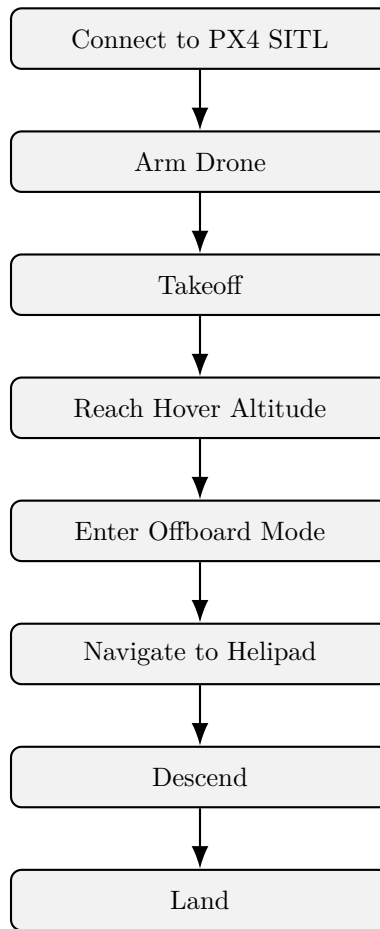


Figure 4: Workflow of the autonomous landing algorithm.

This approach consistently produced stable autonomous flights and successful landings on the designated helipad.

8. Results and Observations

The completed system successfully achieved the objectives of the challenge.

Key observations include:

- Stable hover achieved after PID tuning.
- Oscillations significantly reduced.
- Reliable autonomous takeoff.
- Successful transition into Offboard mode.
- Consistent waypoint navigation.
- Successful autonomous landing on the designated helipad.
- Multiple successful simulation runs were completed before recording the final demonstration.

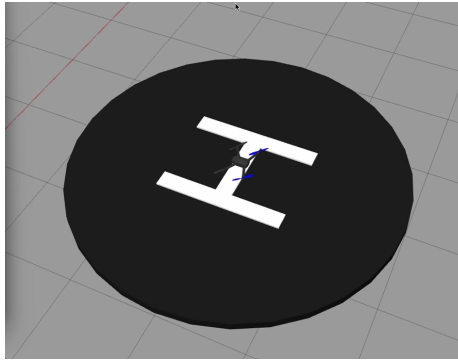


Figure 5: Successful Autonomous Landing

9. Conclusion

This project provided practical experience in multicopter control, PX4 parameter tuning, and autonomous flight using MAVSDK. Through systematic PID tuning, the initially unstable drone was transformed into a stable platform capable of reliable autonomous navigation and precision landing.

The project also demonstrated the importance of iterative controller tuning, proper Offboard mode initialization, and coordinate frame verification when developing autonomous aerial robotics applications. The developed solution successfully demonstrated reliable autonomous takeoff, waypoint navigation, and precision landing within the PX4-Gazebo simulation environment, satisfying the objectives of the challenge.